

What APEX Can Do

A practical view of the Semester Planning Agent

Live demo: <https://technion-semester-planner-agent.vercel.app>

In One Sentence

APEX turns a student's situation into a verified semester plan, then helps revise it when the student changes constraints, asks about courses, or wants a safer path.

Capability Map

What the student needs	What APEX does
Build a semester	Creates a course list with credits, workload, exams, weekly schedule, and a short explanation.
Stay within rules	Checks prerequisites, already-passed courses, unknown courses, duplicate courses, credit range, mandatory progress, schedule clashes, workload, and exam spacing.
Use real course signals	Uses Technion catalog data, schedules, exam dates, CheeseFork difficulty and satisfaction, and grade histogram averages.
Handle retakes	Prioritizes failed retakes and can suggest grade-improvement retakes only after comparing the student's grade, course average, personal average, and reviews.
Answer course questions	Answers focused questions about a course using review data instead of making a full semester plan.
Revise an existing plan	Keeps as much of the old plan as possible while adding, removing, or swapping courses.
Read transcript data	Accepts a transcript PDF and extracts courses and grades for planning.
Keep continuity	Can restore saved sessions through Supabase using a hashed student email.

Example 1: Verified Semester Plan

The agent delivers a plan only after checking it. The student sees the result in a compact plan view with verification status, total credits, workload, course numbers, exams, and course names.

Semester Plan

הנדסת מערכות מידע

✓ Verified · 18.5 credits · workload 46.8/100

00940224 · 4 pts · exam 2027-02-23	מבני נתונים ואלגוריתמים
00940241 · 3 pts · exam 2027-02-03	ניהול מסדי נתונים
00940424 · 3.5 pts · exam 2027-02-07	סטטיסטיקה 1
00940594 · 3.5 pts · exam 2027-02-17	עקרונות הכלכלה למהנדסים
03240521 · 2 pts	יזמות בארגונים-התפתחויות ומגמות
03240675 · 1.5 pts · exam 2027-01-29	ספרדית למתחילים
03940820 · 1 pts	חינוך גופני - תנועה ומחול

Built by my Semester Planning Agent 🤖

A verified 18.5-credit plan with workload score and exam dates.

Example 2: Grade-Improvement Retake Advice

APEX does not blindly retake every low grade. It compares the student's grade with the historical course average, the student's own average, and CheeseFork reviews. Then it explains whether the retake is worth considering.

you

האם אתה ממליץ לי לשפר קורס? לפי הציונים שלי

AGENT

לפי הנתונים, **סטטיסטיקה 1** הוא מועמד סביר לשיפור: קיבלת בו **70**, שזה קצת מתחת למוצע ההיסטורי של **71.8**, אבל הרבה מתחת למוצע האישי הידוע שלך (**86.5**), כך שהציון הזה בולט לרעה ביחס אליך. מצד שני, המשוב על הקורס חזק מאוד — יש **41** חוות דעת, עם שביעות רצון ממוצעת **4.07/5** וקושי **3.49/5**, וביקורות כמו "**מרצה מדהימה באמת**" ו-"**הכי טובה שיש!**" מרמזות שהקורס עצמו סביר ואף אהוד, כך ששיפור יכול להיות כדאי בעיקר אם אתה חושב שתוכל לחזור קרוב יותר לרמתך הרגילה.

backend: <https://technion-semester-palnn>

The agent explains why Statistics 1 is a reasonable grade-improvement retake for this student.

Example 3: Honest Follow-Up Revision

When the student asks for a change, the agent tries to preserve the existing plan. If a requested course creates a real timetable conflict, it says so directly and asks for confirmation before forcing the exception.



Here's your updated plan. I kept 7 of your 8 courses, removed סדנה בסטטיסטיקה לדוקטורנטים, and added 19.5 — סדרות עתיות וחיווי credits total.

Still open: class time overlaps with (11:30-13:30) התנהגות ארגונית in every section combination; only 1 mandatory course(s) in the plan, expected at least 3 - a semester should make real degree progress unless the student asked for fewer.

Note: סדרות עתיות וחיווי was removed from the plan because its class times collide with another course in every available section combination - a clashing week is never delivered. עיבוד שפה טבעית: you asked to add it, but I couldn't - its class times collide with ניהול פרויקטים in every available section combination. Want me to add it anyway despite that? Confirm and I will, no matter what.

Anything you'd like to change? Pick a quick action or reply below.

A follow-up plan revision that keeps most courses, names the remaining conflict, and waits for explicit approval.

Other Things APEX Handles

- Hebrew or English free-text intake.
- Three supported Technion tracks.
- Blocked weekdays and section-level timetable checks.
- Light, normal, and fast pace preferences.
- Exam calendar and study-period pressure.
- Risk report and roadmap to graduation.
- Live offering check before final delivery.
- Required assignment endpoints with prompt, response, and model-step trace.

Limits and Future Work

- The bundled catalog is a snapshot. It will not update automatically when Technion publishes a new catalog unless the pipeline is rerun.
- The system does not register students to courses or access private Technion accounts.
- When constraints conflict, the agent may return the best honest plan and clearly name what is still unresolved.
- Future work could add focused policy retrieval for regulations, forms, and administrative guidance.